

Week 5 In-Class Worksheet

Hypothesis Testing

Name: _____

Date: _____

Course: H524 Introduction to Biostatistics

Topic: Hypothesis Testing for Population Means

Instructions

Work with a partner to solve the following problems. Show all your work, including formulas and calculations. You may use R to verify your answers, but demonstrate your understanding by showing the steps manually first.

1 Problem 1: Two-Sided t-test

A researcher claims that the average body temperature is not 98.6°F. A sample of 25 healthy adults has a mean temperature of 98.4°F with a standard deviation of 0.8°F. Test the claim at the 0.05 significance level.

Given: $n = 25$, $\bar{x} = 98.4^\circ\text{F}$, $s = 0.8^\circ\text{F}$, $\alpha = 0.05$

Part A: State the null and alternative hypotheses.

Part B: Calculate the standard error and t-statistic.

Formula: $SE = \frac{s}{\sqrt{n}}$

Formula: $t = \frac{\bar{x} - \mu_0}{SE}$

Part C: Find the p-value.

Degrees of freedom: $df = n - 1 =$

For a two-sided test, $p\text{-value} = 2 \times P(T < t)$ where t is your test statistic.

(Use R: $2 * pt(t, df=24)$)

Part D: Make a decision and state your conclusion in context.

2 Problem 2: One-Sided t-test (Upper Tail)

A nutritionist claims that a new diet increases average daily caloric expenditure above the baseline of 260 calories. A study of 40 participants following the diet shows a mean expenditure of 285 calories with a standard deviation of 48 calories. Test this claim at the 0.05 level.

Given: $n = 40$, $\bar{x} = 285$, $s = 48$, $\mu_0 = 260$, $\alpha = 0.05$

Part A: State the hypotheses and identify the type of test.

Part B: Calculate the standard error and t-statistic.

Standard error: $SE = \frac{s}{\sqrt{n}} =$

t-statistic: $t = \frac{\bar{x} - \mu_0}{SE} =$

Part C: Find the p-value and make a decision.

Degrees of freedom: $df =$

For a one-sided (upper tail) test: $p\text{-value} = P(T > t)$
(Use R: `1 - pt(t, df)`)

Decision:

Part D: Interpret the results in context. What does this tell you about the diet's effect?

3 Problem 3: Critical Value Approach

Using the data from Problem 2, verify your conclusion using the critical value approach instead of the p-value approach.

Data: $n = 40$, $\bar{x} = 285$, $s = 48$, $t = [\text{your calculated value}]$, $\alpha = 0.05$

Part A: Find the critical value for this one-sided test.

For a one-sided (upper tail) test with $\alpha = 0.05$ and $df = 39$:

(Use R: `qt(0.95, df=39)`)

Part B: Compare the t-statistic to the critical value and make a decision.

Part C: Compare this conclusion to the p-value approach from Problem 2. Do they agree?

4 Problem 4: Type I and Type II Errors

A pharmaceutical company is testing a new treatment that they hope is more effective than the standard treatment.

Part A: Define a Type I error in the context of this study. What would be the consequence?

Part B: Define a Type II error in the context of this study. What would be the consequence?

Part C: Which type of error do you think is more serious in medical research? Explain your reasoning.

5 Problem 5: Statistical Power and Sample Size

Answer the following conceptual questions about statistical power.

Part A: How does increasing the significance level (e.g., from 0.05 to 0.10) affect the probability of Type I and Type II errors? What happens to statistical power?

Part B: How does increasing sample size affect Type I error, Type II error, and statistical power? Explain the mechanism.

6 Problem 6: Comprehensive Integration Problem

A pharmaceutical company is testing whether a new sleep medication reduces the average sleep time below the standard 8 hours. They conduct a study with 50 patients and find a mean sleep time of 7.8 hours with a standard deviation of 1.2 hours. Test the hypothesis at the $\alpha = 0.01$ level.

Given: $n = 50$, $\bar{x} = 7.8$ hours, $s = 1.2$ hours, $\alpha = 0.01$

Part A: State the null and alternative hypotheses. Is this a one-sided or two-sided test?

Part B: Calculate the standard error.

Formula: $SE = \frac{s}{\sqrt{n}}$

Part C: Calculate the t-statistic.

Formula: $t = \frac{\bar{x} - \mu_0}{SE}$

Part D: Find the p-value.

Degrees of freedom: $df =$

For a one-sided (lower tail) test: $p\text{-value} = P(T < t)$
(Use R: `pt(t, df)`)

Part E: Make a decision at the $\alpha = 0.01$ level.

Part F: Write a complete interpretation of your results in the context of the sleep study.

Part G: Verify your conclusion using the critical value approach.

For a one-sided (lower tail) test with $\alpha = 0.01$ and $df = 49$:
(Use R: `qt(0.01, df=49)`)

Compare your t-statistic to the critical value:

Reflection Questions

1. Explain the difference between statistical significance and practical significance. Can a result be statistically significant but not practically important?

2. Why do we typically use $\alpha = 0.05$ as the significance level? What would happen if we used $\alpha = 0.10$ or $\alpha = 0.01$ instead?

3. When would you use a one-sided test versus a two-sided test? Give an example of each.

4. What is the relationship between confidence intervals and hypothesis testing?